DEVAK SHARMA

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EDUCATION

University of Central Florida *Orlando, FL | May 2026*

B.S. Computer Engineering | Minor: Intelligent Robotic Systems | GPA: 3.4

Relevant Coursework: Embedded Systems, Computer Architecture, Linear Control Systems, Operating Systems, Data Structures & Algorithms, Artificial Intelligence, Robotic Systems

EXPERIENCE

Embedded Systems Engineering Intern *January 2026–Present*

[AMROK] — Orlando, FL

- Developed Python scripts to automate Rendezvous, Proximity Operations & Docking (RPOD) missions in Low Earth Orbit (LEO), enabling autonomous satellite guidance and maneuver sequencing.
- Implemented computer vision pipelines using Python and OpenCV for real-time image processing tasks including object detection, feature extraction, and spatial tracking in support of autonomous docking operations.
- Contributing to embedded sensor hardware development for satellite payloads, integrating and validating peripherals on a Raspberry Pi 5 platform including I/O configuration, driver setup, and sensor communication.
- Collaborate with engineers on hardware-software co-design, debugging sensor interfaces and validating system behavior in a space-systems development environment.

PROJECTS

Automated Turret — Senior Design *2025–2026*

- Built an automated targeting system using the ESP32-S3 micro-controller with a PID controller for stable pan/tilt tracking under a heavy payload, mitigating electrical noise from ESC motor drivers and solenoid actuation.
- Deployed a YOLO model for real-time color and object detection, feeding recognized targets into a PID tracking controller capable of following moving targets up to ~5°/sec; achieved lock-on in ~2 seconds and ~95% hit accuracy on stationary targets under controlled lighting conditions.
- Managed hardware-software co-design across a multi-person team through formal prototyping and Critical Design Review (CDR) phases, presenting to a technical panel for peer review and Q&A.

Custom Printed Circuit Board (PCB) *2025*

- Designed a two-layer PCB in Autodesk Fusion featuring onboard 3.3V and 5V switching regulators, with careful component placement and trace routing to minimize noise for analog sensor readings.
- Programmed a TI MSP430G2553 micro-controller in C using Code Composer Studio for real-time ultrasonic distance acquisition, signal processing, and dynamic LED display output.
- Validated the assembled board against design specifications using an oscilloscope and multimeter; resolved a voltage ripple issue through decoupling capacitor placement.

Self-Balancing Robot *2024*

- Developed embedded firmware for a two-wheeled autonomous robot using a PID controller fed by IMU sensor data, achieving stable upright balance within ±2° tilt tolerance.
- Applied a complementary filter to blend accelerometer and gyroscope readings, correcting for drift and dynamic motion errors to maintain reliable orientation estimates during movement.
- Performed full hardware bring-up including frame assembly, motor driver wiring, soldering, and sensor calibration; tuned PID gains empirically to eliminate oscillation under load.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, TypeScript, MATLAB

Embedded / Hardware: Raspberry Pi, TI MSP430, ESP32, PCB Design, UART/SPI/I2C, Multimeter, Oscilloscope, Soldering

Software & Tools: Git, Linux, Code Composer Studio, React, Node.js, Autodesk Fusion, VS Code, ROS, LTSpice, MATLAB

Concepts: RPOD Autonomy, Firmware Development, PID Control, Hardware-Software Co-design, UX/UI Design